ATF1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes ATF1 gene fusions from the msk_impact_50k_2026 study, covering 30 fusion events. 21/25 fusions with known frame status (84.0%) were in-frame. 27/30 fusions (90.0%) retained the bZIP transcription factor domain. Domain retention was tested with Fisher's exact test (p=0.704212, not statistically significant at alpha=0.05). No literature benchmark is configured for ATF1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 30 analyzed fusion events, identifying 2 distinct fusion partner genes. The most frequent partner was EWSR1, observed in 29 events.

Domain retention

bZIP transcription factor domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 19/21 (90.5%) of in-frame fusions retained the domain, p=0.704212 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.158416.

Domain disruption

Disruption of the pKID domain was tested with Fisher's exact test comparing in-frame fusions against all others; 18/19 (94.7%) of in-frame fusions disrupted the domain, p=0.547619 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.99998e-05.

Junction concentration

64% of 25 fusions with this gene as the 3' partner join at exon 4, against 24% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.00029997, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 28 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 31 aa (exon 2; label-permutation p=0.036963). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -12 aa from the nearest configured domain boundary at 43 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=25) and "not_retained" (n=3). For Frame_status, retained: 17/21 (81.0%, 95% CI 60.0%-92.3%); not_retained: 2/2 (100.0%, 95% CI 34.2%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of 44.36 vs 22.6667, p=0.154983 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score produced results for this run (2 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for ATF1 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (18 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 3 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: ATF1 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: ATF1 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (23 row(s) in window_scan, 10 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EWSR1	29	0.9666666666666667	0.08002013911267178	0.46989743472591233	0.0	0.5095251155661108	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.4549370332799615	1
ERBB3	1	0.03333333333333333	0.08002013911267178	0.46989743472591233	0.33333333333333337	0.5095251155661108	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2249370332799615	2

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1	1	24	1	2	0.08333333333333333	0.20634920634920637	0.6853971971467449
31	1	24	2	1	0.020833333333333332	0.0231990231990232	1.6345303008075895
32	2	23	2	1	0.043478260869565216	0.045177045177045176	1.3450821776934234
65	18	7	3	0	0.0	0.5512820512820513	0.2586261471108938
110	24	1	3	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

18	8
1	1

frame_domain_contingency_tables_by_tumor_type

stratum	table
AFH	[2, 0], [0, 0]
AN	[1, 0], [0, 0]
CCS	[11, 5], [0, 0]

stratum	table
LUAD	[0, 1], [0, 0]
MUCC	[1, 0], [0, 0]
MYEC	[0, 0], [1, 0]
OPHSC	[1, 0], [0, 0]
OSACA	[1, 0], [0, 1]
PEMESO	[0, 1], [0, 0]
PLEMESO	[0, 1], [0, 0]
SARCNOS	[1, 0], [0, 0]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf02173	pKID domain	28	2	17	9	0.29903846153846153

domain_retention tables

frame_domain_contingency_table

17	8
2	1

frame_domain_contingency_tables_by_tumor_type

stratum	table
AFH	[2, 0], [0, 0]
AN	[1, 0], [0, 0]
CCS	[10, 5], [1, 0]
LUAD	[0, 0], [0, 1]
MUCC	[1, 0], [0, 0]
MYEC	[1, 0], [0, 0]
OPHSC	[1, 0], [0, 0]
OSACA	[0, 1], [1, 0]
PEMESO	[0, 1], [0, 0]
PLEMESO	[0, 1], [0, 0]
SARCNOS	[1, 0], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	bZIP transcription factor	28	25	0	3	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
6	25	28	0.8928571428571429

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ERBB3	1	1	1	1
EWSR1	29	29	29	27

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
12	51199000	51200000	4	EVT-P-0011423-T01-IM5-2, EVT-P-0026677-T01-IM6-13, EVT-P-0027614-T02-IM6-15, EVT-P-0049830-T02-IM6-25
12	51200000	51201000	3	EVT-P-0031637-T01-IM6-17, EVT-P-0032329-T02-IM6-18, EVT-P-0051953-T02-IM6-28
12	51193000	51194000	2	EVT-P-0000277-T02-IM6-6, EVT-P-0064738-T01-IM7-29
12	51201000	51202000	2	EVT-P-0021629-T01-IM6-11, EVT-P-0047115-T01-IM6-23
12	51204000	51205000	2	EVT-P-0026818-T01-IM6-14, EVT-P-0029754-T01-IM6-16
12	51205000	51206000	2	EVT-P-0044247-T01-IM6-22, EVT-P-0068570-T01-IM7-30
12	51207000	51208000	2	EVT-P-0013282-T01-IM5-9, EVT-P-0041891-T02-IM6-21
12	51160000	51161000	1	EVT-P-0008210-T01-IM5-5
12	51167000	51168000	1	EVT-P-0029943-T01-IM6-1

chromosome	bin_start	bin_end	n_events	event_ids
12	51175000	51176000	1	EVT-P-0001872-T01-IM3-4
12	51184000	51185000	1	EVT-P-0025162-T01-IM6-12
12	51190000	51191000	1	EVT-P-0019448-T01-IM6-10
12	51191000	51192000	1	EVT-P-0039378-T02-IM6-20
12	51195000	51196000	1	EVT-P-0001214-T01-IM3-3
12	51197000	51198000	1	EVT-P-0032409-T02-IM6-19
12	51202000	51203000	1	EVT-P-0012694-T01-IM5-8
12	51203000	51204000	1	EVT-P-0051365-T01-IM6-27
12	51206000	51207000	1	EVT-P-0012497-T02-IM5-7

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
65	17	17	12207	False
110	6	6	3246	False
1	2	2	6605	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	4	16	25	0.64	0.23895409195685455	4.627593868922764
three_prime	5	7	25	0.28	0.08099452763456565	2.526985467897946
five_prime	2	2	3	0.6666666666666666	0.5563336911192834	0.2334808318299226
five_prime	3	1	3	0.3333333333333333	0.23837342295578118	0.2532106107955504
three_prime	2	1	25	0.04	0.2806038957611163	0.00011535702031665062
three_prime	3	1	25	0.04	0.2775069944220584	0.0001284368674732223

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-168	32	200	4	24	2	23	0.043478260869565216	0.045177045177045176	1.3450821776934234	EVT-P-0001872-T01-IM3-4, EVT-P-0008210-T01-IM5-5, EVT-P-0025162-T01-IM6-12, EVT-P-0029943-T01-IM6-1

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-135	65	200	21	7	18	7	0.0	0.5512820512820513	0.2586261471108938	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0001872-T01-IM3-4, EVT-P-0008210-T01-IM5-5, EVT-P-0011423-T01-IM5-2, ... (21 total)
-68	32	100	4	24	2	23	0.043478260869565216	0.045177045177045176	1.3450821776934234	EVT-P-0001872-T01-IM3-4, EVT-P-0008210-T01-IM5-5, EVT-P-0025162-T01-IM6-12, EVT-P-0029943-T01-IM6-1
-35	65	100	21	7	18	7	0.0	0.5512820512820513	0.2586261471108938	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0001872-T01-IM3-4, EVT-P-0008210-T01-IM5-5, EVT-P-0011423-T01-IM5-2, ... (21 total)
-18	32	50	4	24	2	23	0.043478260869565216	0.045177045177045176	1.3450821776934234	EVT-P-0001872-T01-IM3-4, EVT-P-0008210-T01-IM5-5, EVT-P-0025162-T01-IM6-12, EVT-P-0029943-T01-IM6-1
1	51	50	4	24	2	23	0.043478260869565216	0.045177045177045176	1.3450821776934234	EVT-P-0001872-T01-IM3-4, EVT-P-0008210-T01-IM5-5, EVT-P-0025162-T01-IM6-12, EVT-P-0029943-T01-IM6-1
1	101	100	21	7	18	7	0.0	0.5512820512820513	0.2586261471108938	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0001872-T01-IM3-4, EVT-P-0008210-T01-IM5-5, EVT-P-0011423-T01-IM5-2, ... (21 total)
15	65	50	19	9	17	8	1.0625	1.0	-0.0	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0001872-T01-IM3-4, EVT-P-0011423-T01-IM5-2, EVT-P-0012694-T01-IM5-8, ... (19 total)
31	81	50	19	9	17	8	1.0625	1.0	-0.0	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0001872-T01-IM3-4, EVT-P-0011423-T01-IM5-2, EVT-P-0012694-T01-IM5-8, ... (19 total)
32	82	50	18	10	17	8	4.25	0.2838827838827839	0.5468609445344458	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0001872-T01-IM3-4, EVT-P-0011423-T01-IM5-2, EVT-P-0012694-T01-IM5-8, ... (18 total)
32	132	100	24	4	23	2	23.0	0.04517704517704517	1.3450821776934236	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0001872-T01-IM3-4, EVT-P-0011423-T01-IM5-2, EVT-P-0012497-T02-IM5-7, ... (24 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
40	65	25	17	11	16	9	3.5555555555555554	0.5433455433455432	0.264923890779487	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0011423-T01-IM5-2, EVT-P-0012694-T01-IM5-8, EVT-P-0019448-T01-IM6-10, ... (17 total)
60	110	50	23	5	22	3	14.666666666666666	0.07326007326007326	1.1351326513767748	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0011423-T01-IM5-2, EVT-P-0012497-T02-IM5-7, EVT-P-0012694-T01-IM5-8, ... (23 total)
63	163	100	24	4	23	2	23.0	0.04517704517704517	1.3450821776934236	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0011423-T01-IM5-2, EVT-P-0012497-T02-IM5-7, EVT-P-0012694-T01-IM5-8, ... (24 total)
65	90	25	17	11	16	9	3.5555555555555554	0.5433455433455432	0.264923890779487	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0011423-T01-IM5-2, EVT-P-0012694-T01-IM5-8, EVT-P-0019448-T01-IM6-10, ... (17 total)
65	115	50	23	5	22	3	14.666666666666666	0.07326007326007326	1.1351326513767748	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0011423-T01-IM5-2, EVT-P-0012497-T02-IM5-7, EVT-P-0012694-T01-IM5-8, ... (23 total)
65	165	100	24	4	23	2	23.0	0.04517704517704517	1.3450821776934236	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0011423-T01-IM5-2, EVT-P-0012497-T02-IM5-7, EVT-P-0012694-T01-IM5-8, ... (24 total)
65	265	200	24	4	23	2	23.0	0.04517704517704517	1.3450821776934236	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0011423-T01-IM5-2, EVT-P-0012497-T02-IM5-7, EVT-P-0012694-T01-IM5-8, ... (24 total)
85	110	25	6	22	6	19	None	1.0	-0.0	EVT-P-0012497-T02-IM5-7, EVT-P-0026818-T01-IM6-14, EVT-P-0029754-T01-IM6-16, EVT-P-0041891-T02-IM6-21, EVT-P-0044247-T01-IM6-22, ... (6 total)
110	135	25	6	22	6	19	None	1.0	-0.0	EVT-P-0012497-T02-IM5-7, EVT-P-0026818-T01-IM6-14, EVT-P-0029754-T01-IM6-16, EVT-P-0041891-T02-IM6-21, EVT-P-0044247-T01-IM6-22, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
110	160	50	6	22	6	19	None	1.0	-0.0	EVT-P-0012497-T02-IM5-7, EVT-P-0026818-T01-IM6-14, EVT-P-0029754-T01-IM6-16, EVT-P-0041891-T02-IM6-21, EVT-P-0044247-T01-IM6-22, ... (6 total)
110	210	100	7	21	7	18	None	0.5512820512820512	0.25862614711089393	EVT-P-0012497-T02-IM5-7, EVT-P-0013282-T01-IM5-9, EVT-P-0026818-T01-IM6-14, EVT-P-0029754-T01-IM6-16, EVT-P-0041891-T02-IM6-21, ... (7 total)
110	310	200	7	21	7	18	None	0.5512820512820512	0.25862614711089393	EVT-P-0012497-T02-IM5-7, EVT-P-0013282-T01-IM5-9, EVT-P-0026818-T01-IM6-14, EVT-P-0029754-T01-IM6-16, EVT-P-0041891-T02-IM6-21, ... (7 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
32	132	100	24	4	23	2	23.0	0.04517704517704517	1.3450821776934236	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0001872-T01-IM3-4, EVT-P-0011423-T01-IM5-2, EVT-P-0012497-T02-IM5-7, ... (24 total)
63	163	100	24	4	23	2	23.0	0.04517704517704517	1.3450821776934236	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0011423-T01-IM5-2, EVT-P-0012497-T02-IM5-7, EVT-P-0012694-T01-IM5-8, ... (24 total)
-18	32	50	4	24	2	23	0.043478260869565216	0.045177045177045176	1.3450821776934234	EVT-P-0001872-T01-IM3-4, EVT-P-0008210-T01-IM5-5, EVT-P-0025162-T01-IM6-12, EVT-P-0029943-T01-IM6-1
60	110	50	23	5	22	3	14.666666666666666	0.07326007326007326	1.1351326513767748	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0011423-T01-IM5-2, EVT-P-0012497-T02-IM5-7, EVT-P-0012694-T01-IM5-8, ... (23 total)
32	82	50	18	10	17	8	4.25	0.2838827838827839	0.5468609445344458	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0001872-T01-IM3-4, EVT-P-0011423-T01-IM5-2, EVT-P-0012694-T01-IM5-8, ... (18 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
40	65	25	17	11	16	9	3.5555555555555554	0.5433455433455432	0.264923890779487	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0011423-T01-IM5-2, EVT-P-0012694-T01-IM5-8, EVT-P-0019448-T01-IM6-10, ... (17 total)
110	210	100	7	21	7	18	None	0.5512820512820512	0.25862614711089393	EVT-P-0012497-T02-IM5-7, EVT-P-0013282-T01-IM5-9, EVT-P-0026818-T01-IM6-14, EVT-P-0029754-T01-IM6-16, EVT-P-0041891-T02-IM6-21, ... (7 total)
-35	65	100	21	7	18	7	0.0	0.5512820512820513	0.2586261471108938	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0001872-T01-IM3-4, EVT-P-0008210-T01-IM5-5, EVT-P-0011423-T01-IM5-2, ... (21 total)
85	110	25	6	22	6	19	None	1.0	-0.0	EVT-P-0012497-T02-IM5-7, EVT-P-0026818-T01-IM6-14, EVT-P-0029754-T01-IM6-16, EVT-P-0041891-T02-IM6-21, EVT-P-0044247-T01-IM6-22, ... (6 total)
15	65	50	19	9	17	8	1.0625	1.0	-0.0	EVT-P-0000277-T02-IM6-6, EVT-P-0001214-T01-IM3-3, EVT-P-0001872-T01-IM3-4, EVT-P-0011423-T01-IM5-2, EVT-P-0012694-T01-IM5-8, ... (19 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029943-T01-IM6-1	P-0029943-T01-IM6	P-0029943	Non-Small Cell Lung Cancer	LUAD	ATF1-ERBB3	ERBB3	unknown	five_prime	51167430	2	1	True			breakpoint lies entirely upstream of the first coding exon (zero coding bases retained on this side); no residue to compare	lost	0.0	False		bZIP transcription factor; pKID domain		ATF1 (NM_005171) - ERBB3 (NM_001982) rearrangement: c.-6-6493:ATF1_c.1818:ERBB3del | Protein Fusion: mid-exon {ATF1:ERBB3}	NA
EVT-P-0011423-T01-IM5-2	P-0011423-T01-IM5	P-0011423	Soft Tissue Sarcoma	CCS	ATF1-EWSR1	EWSR1	in-frame	five_prime	51199798	3	65	True	64	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		bZIP transcription factor	pKID domain	ATF1 (NM_005171) - EWSR1 (NM_013986) fusion (ATF1 exons 1-3 with EWSR1 exons 10-18): t(12;22)(q13.12;q12.2)(chr12:g.51199798::chr22:g.29686903) | Protein fusion: in frame (ATF1-EWSR1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001214-T01-IM3-3	P-0001214-T01-IM3	P-0001214	Salivary Gland Cancer	MUCC	EWSR1-ATF1	EWSR1	in-frame	three_prime	51195854	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_005243) - ATF1 (NM_005171) Reciprocal Translocation: t(22;12)(q12.2;q13.12)(chr22:c.974+915::chr12:c.194+6062) | Protein fusion: in frame (EWSR1-ATF1)	NA
EVT-P-0001872-T01-IM3-4	P-0001872-T01-IM3	P-0001872	Salivary Gland Cancer	MYEC	EWSR1-ATF1	EWSR1	in-frame	three_prime	51175004	3	32	True	32	True	exact match	retained	1.0	False	bZIP transcription factor; pKID domain			NA | Protein fusion: in frame (EWSR1-ATF1)	NA
EVT-P-0008210-T01-IM5-5	P-0008210-T01-IM5	P-0008210	Salivary Gland Cancer	OSACA	EWSR1-ATF1	EWSR1	unknown	three_prime	51160825	2	1	True	1	True	exact match	retained	1.0	False	bZIP transcription factor; pKID domain			EWSR1 (NM_013986) - ATF1 (NM_005171) rearrangement: t(12;22)(q13;q12)(chr12:g.51160825:: chr22:g.29691427) | Transcript fusion (EWSR1-ATF1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000277-T02-IM6-6	P-0000277-T02-IM6	P-0000277	Head and Neck Cancer	OPHSC	EWSR1-ATF1	EWSR1	in-frame	three_prime	51193042	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.12;q12.2)(chr12:g.51193042::chr22:g.29686107) | Protein Fusion: in frame {EWSR1:ATF1}	NA
EVT-P-0012497-T02-IM5-7	P-0012497-T02-IM5	P-0012497	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51206235	5	110	True	110	True	exact match	retained	1.0	False	bZIP transcription factor	pKID domain		EWSR1 (NM_005243) - ATF1 (NM_005171) Fusion (EWSR1 exons 1-9 fused with ATF1 exons 5-7) : t(12;22)(q13;q12)(chr12:g.51206235::chr22:g.29687613) | Protein fusion: in frame (EWSR1-ATF1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012694-T01-IM5-8	P-0012694-T01-IM5	P-0012694	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	unknown	three_prime	51202318	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion (EWSR1 exons 1-10 with ATF1 exons 4-7): t(12;22)(q13.12;q12.2)(chr12:g.51202318::chr22:g.29687578) | Protein fusion: mid-exon (EWSR1-ATF1)	NA
EVT-P-0013282-T01-IM5-9	P-0013282-T01-IM5	P-0013282	Mesothelioma	PEMESO	EWSR1-ATF1	EWSR1	unknown	three_prime	51207953	5	163	False	163	True	exact match	retained	1.0	False	bZIP transcription factor	pKID domain		EWSR1 (NM_013986) - ATF1 (NM_005171) fusion (EWSR1 exons 1-14 with ATF1 exons 5-7): t(12;22)(q13.12;q12.2)(chr12:g.51207953)(chr22:g.29694628) | Protein fusion: mid-exon (EWSR1-ATF1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019448-T01-IM6-10	P-0019448-T01-IM6	P-0019448	Angiomatoid Fibrous Histiocytoma	AFH	EWSR1-ATF1	EWSR1	in-frame	three_prime	51190862	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion (EWSR1 exons 1-9 fused to ATF1 exons 4-7): t(12;22)(q13.12;q12.2)(chr12:g.51190862::chr22:g.96871109) | Protein Fusion: in frame {EWSR1:ATF1}	NA
EVT-P-0021629-T01-IM6-11	P-0021629-T01-IM6	P-0021629	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	out-of-frame	three_prime	51201619	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion (EWSR1 exons 1-10 fused in frame with ATF1 exons 4-7): t(12;22)(q13.12;q12.2)(chr12:g.51201619::chr22:g.29687948) | Protein Fusion: out of frame {EWSR1:ATF1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025162-T01-IM6-12	P-0025162-T01-IM6	P-0025162	Salivary Gland Cancer	OSACA	EWSR1-ATF1	EWSR1	in-frame	five_prime	51184357	2	31	True	31	True	exact match	lost	0.0	False		bZIP transcription factor; pKID domain		ATF1 (NM_005171) - EWSR1 (NM_013986) rearrangement: t(12;22)(q13.12;q12.2)(chr12:g.51184357::chr22:g.29691768) | Protein Fusion: in frame {ATF1:EWSR1}	NA
EVT-P-0026677-T01-IM6-13	P-0026677-T01-IM6	P-0026677	Skin Cancer, Non-Melanoma	AN	EWSR1-ATF1	EWSR1	in-frame	three_prime	51199267	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion (EWSR1 exons 1-9 fused in-frame to ATF1 exons 4-7): t(12;22)(12q13.12;22q12.2)(chr12:g.51199267::chr22:g.29686620) | Protein Fusion: in frame {EWSR1:ATF1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026818-T01-IM6-14	P-0026818-T01-IM6	P-0026818	Soft Tissue Sarcoma	SARCNOS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51204029	5	110	True	110	True	exact match	retained	1.0	False	bZIP transcription factor	pKID domain		EWSR1 (NM_013986) - ATF1 (NM_005171) fusion (EWSR1 exons 1-8 with ATF1 exons 5-7): t(12;22)(q13.12;q12.2)(chr12:g.51207953)(chr22:g.29694628) | Protein Fusion: in frame {EWSR1:ATF1}	NA
EVT-P-0027614-T02-IM6-15	P-0027614-T02-IM6	P-0027614	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51199959	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion (EWSR1 exons 1-9 with ATF1 exons 4-7): t(12;22)(q13.12;q12.2)(chr12:g.51199959)(chr22:g.29687072) | Protein Fusion: in frame {EWSR1:ATF1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029754-T01-IM6-16	P-0029754-T01-IM6	P-0029754	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51204764	5	110	True	110	True	exact match	retained	1.0	False	bZIP transcription factor	pKID domain		EWSR1 (NM_013986) - ATF1 (NM_005171) fusion (EWSR1 exons 1-8 fused to ATF1 exons 5-7): t(12;22)(q13.12;q12.2)(chr12:g.51204764::chr22:g.29683656) | Protein Fusion: in frame {EWSR1:ATF1}	NA
EVT-P-0031637-T01-IM6-17	P-0031637-T01-IM6	P-0031637	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51200430	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.13;q12.2)(chr12:g.51200430::chr22:g.29686066) | Protein Fusion: in frame {EWSR1:ATF1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032329-T02-IM6-18	P-0032329-T02-IM6	P-0032329	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51200627	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_005243) - ATF1 (NM_005171) Fusion (EWSR1 exons 1-8 fused with ATF1 exons 4-7) : t(12;22)(q13.13;q12)(chr12:g.51200627::chr22:g.29686632) | Protein Fusion: in frame {EWSR1:ATF1}	NA
EVT-P-0032409-T02-IM6-19	P-0032409-T02-IM6	P-0032409	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	unknown	three_prime	51197338	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion (EWSR1 exons 1-11 fused to ATF1 exons 4-7): t(12;22)(q13.12;q12.2)(chr12:g.51197338::chr22:g.29688134) | Protein Fusion: mid-exon {EWSR1:ATF1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039378-T02-IM6-20	P-0039378-T02-IM6	P-0039378	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51191610	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) rearrangement: t(12;22)(q13.12;q12.2)(chr12:g.51191610::chr22:g.29686513) | Protein Fusion: in frame {EWSR1:ATF1}	NA
EVT-P-0041891-T02-IM6-21	P-0041891-T02-IM6	P-0041891	Angiomatoid Fibrous Histiocytoma	AFH	EWSR1-ATF1	EWSR1	in-frame	three_prime	51207275	5	110	True	110	True	exact match	retained	1.0	False	bZIP transcription factor	pKID domain		EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.12;q12.2)(chr12:g.51207275::chr22:g.29684261) | Protein Fusion: in frame {EWSR1:ATF1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0044247-T01-IM6-22	P-0044247-T01-IM6	P-0044247	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51205619	5	110	True	110	True	exact match	retained	1.0	False	bZIP transcription factor	pKID domain		EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.12;q12.2)(chr12:g.51205619::chr22:g.29684238) | Protein Fusion: in frame {EWSR1:ATF1}	NA
EVT-P-0047115-T01-IM6-23	P-0047115-T01-IM6	P-0047115	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51201743	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.12;q12.2)(chr12:g.51201743::chr22:g.29685486) | Protein Fusion: in frame {EWSR1:ATF1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047115-T02-IM6-24	P-0047115-T02-IM6	P-0047115	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51201746	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.12;q12.2)(chr12:g.51201746::chr22:g.29685485) | Protein Fusion: in frame {EWSR1:ATF1}	NA
EVT-P-0049830-T02-IM6-25	P-0049830-T02-IM6	P-0049830	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51199670	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.12;q12.2)(chr12:g.51199670::chr22:g.29685150) | Protein Fusion: in frame {EWSR1:ATF1}	NA

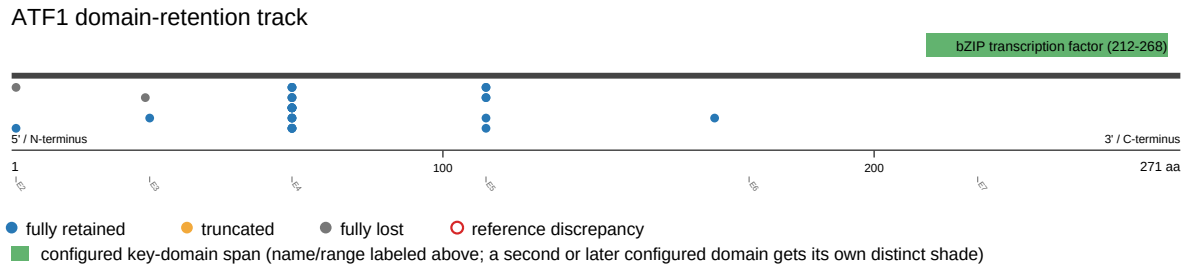
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049830-T03-IM6-26	P-0049830-T03-IM6	P-0049830	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51199670	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.12;q12.2)(chr12:g.51199670::chr22:g.29685150) | Protein Fusion: in frame {EWSR1:ATF1}	NA
EVT-P-0051365-T01-IM6-27	P-0051365-T01-IM6	P-0051365	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	out-of-frame	three_prime	51203069	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.12;q12.2)(chr12:g.51203069::chr22:g.29687795) | Protein Fusion: out of frame {EWSR1:ATF1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051953-T02-IM6-28	P-0051953-T02-IM6	P-0051953	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	out-of-frame	three_prime	51200433	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.12;q12.2)(chr12:g.51200433::chr22:g.29687604) | Protein Fusion: out of frame {EWSR1:ATF1}	NA
EVT-P-0064738-T01-IM7-29	P-0064738-T01-IM7	P-0064738	Soft Tissue Sarcoma	CCS	EWSR1-ATF1	EWSR1	in-frame	three_prime	51193890	4	65	True	65	True	exact match	retained	1.0	False	bZIP transcription factor		pKID domain	EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.12;q12.2)(chr12:g.51193890::chr22:g.29687328) | Protein Fusion: in frame {EWSR1:ATF1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068570-T01-IM7-30	P-0068570-T01-IM7	P-0068570	Mesothelioma	PLEMESO	EWSR1-ATF1	EWSR1	out-of-frame	three_prime	51205611	5	110	True	110	True	exact match	retained	1.0	False	bZIP transcription factor	pKID domain		EWSR1 (NM_013986) - ATF1 (NM_005171) fusion: t(12;22)(q13.12;q12.2)(chr12:g.51205611::chr22:g.29686465) | Protein Fusion: out of frame {EWSR1:ATF1}	NA

Figures

domain retention outliers



fusion schematic

